10.4.2. Factors influencing the antigenicity of antigens

Ability of an antigen to induce an immune response is known as antigenicity. The factors that influence antigenicity are mainly of two types i.e. factors contributed by antigens and factors contributed by host cells.

Factors contributed by Antigens:

The factors are: size, foreignness, chemical nature, complexity, heterogeneity and susceptibility for antigen processing.

Size

Antigens with greater than 10,000 Daltons molecular weight are found to be effective antigens. For example: Hemoglobin is more antigenic than penicillin.

Foreignness (Alien)

It refers to the distance in the phylogenetic tree i.e. evolutionary distance. Highly distant organisms are found to have more antigenicity than less distant ones. For example: Protein from gorilla is found to have less antigenicity than the protein from fish to humans because fish is more evolutionarily distant from gorilla.

Chemical Nature

Depending on the chemical nature, antigenicity varies.; For example, proteins have greater antigenicity than carbohydrates, lipids and nucleicacids. Solubility also plays an important role. Less soluble antigens have more antigenicity.

Complexity

As the complexity of the antigens increase antigenicity also increases. For example, Primary structure of proteins has lesser antigenicity than tertiary structure of the same protein.

Heterogeneity

In the case of multimeric proteins, heteromultimeric proteins have more antigenicity than homomultimeric proteins. This is mainly because of presence of the different types of epitopes in heteromultimeric proteins.

Susceptibility for Antigen Processing

Those antigens which are found to be easily processed by Antigen Presenting Cells (APCs), have greater antigenicity than those antigens which are not easily processed. For example: Horse RBCs have greater antigenicity than asbestos.

Factors contributed by Host

These factors include route of entry, genotype and dose.

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